

Solve the Linear Euqations

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Stay Hungry, Stay Foolish
---Steve · Jobs

1. Vectors and Linear Equations

First we give two equations :

$$\begin{cases} x - 2y = 1 \\ 3x - 2y = 11 \end{cases}$$

- we begin a row at a time. The first equation $x - 2y = 1$ is a straight line in the xy plane. the point $(1, 0)$ is on the line because it solves that equation. the point $(3, 1)$ also on the plane, that, we have the “**Row picture**”

1.1. Row picture

Definition : The row picture shows two lines meeting at a single point. (this point is the solution to this Equations.)

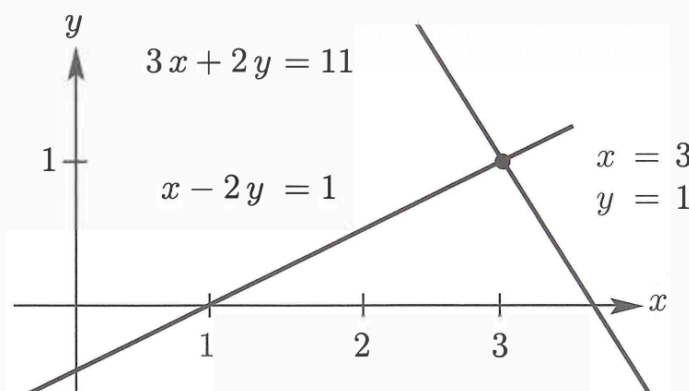


Figure 2.1: Row picture: The point $(3, 1)$ where the lines meet solves both equations

1.2. Column picture

Definition : The columns picture combines the column vectors on the left side to produce the vector $\vec{b}$ on the right side :

$$x \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix} + y \cdot \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \vec{b}$$

so, the problem is “To find the combination of those vectors that equals the vector on the right.”

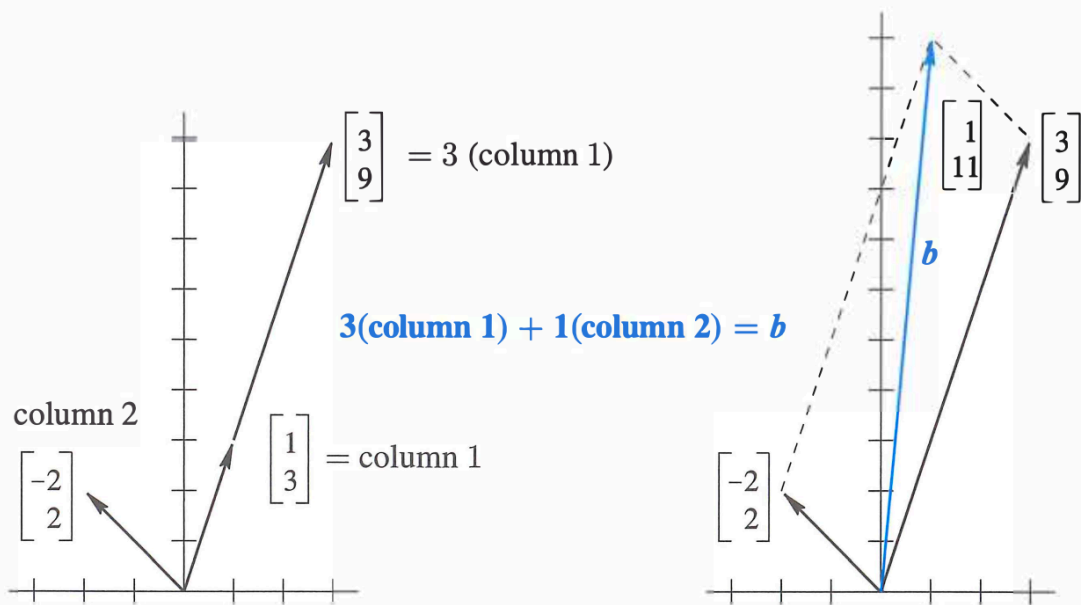


Figure 2.2: *Column picture*: A combination of columns produces the right side (1, 11).

1.3. Coefficient Matrix :

Definition : The Coefficient Matrix on the left side of the equations is the 2 by 2 matrix A :

$$A = \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix}$$

This is a typical of linear algebra : a matrix multiply another matrix :

$$\begin{aligned} A\vec{x} &= \vec{b} \\ \Rightarrow \begin{bmatrix} 1 & -2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 1 \\ 11 \end{bmatrix} \end{aligned}$$

1.4. The Equations in Three Unknowns

If the three Unknowns are

$$xyz$$

, we have the three equations :

$$\begin{cases} x + 2y + 3z = 6 \\ 2x + 5y + 2z = 4 \\ 6x - 3y + z = 2 \end{cases}$$

So, the Row picture and the Column picture will be changed :

- **Row** : The row picture shows three planes meeting at a single point.
- **Column** : The column picture combines three columns to produce $\vec{b} = (6, 4, 2)$

with the Row picture, each equation produces a plane in 3-dimensional space.

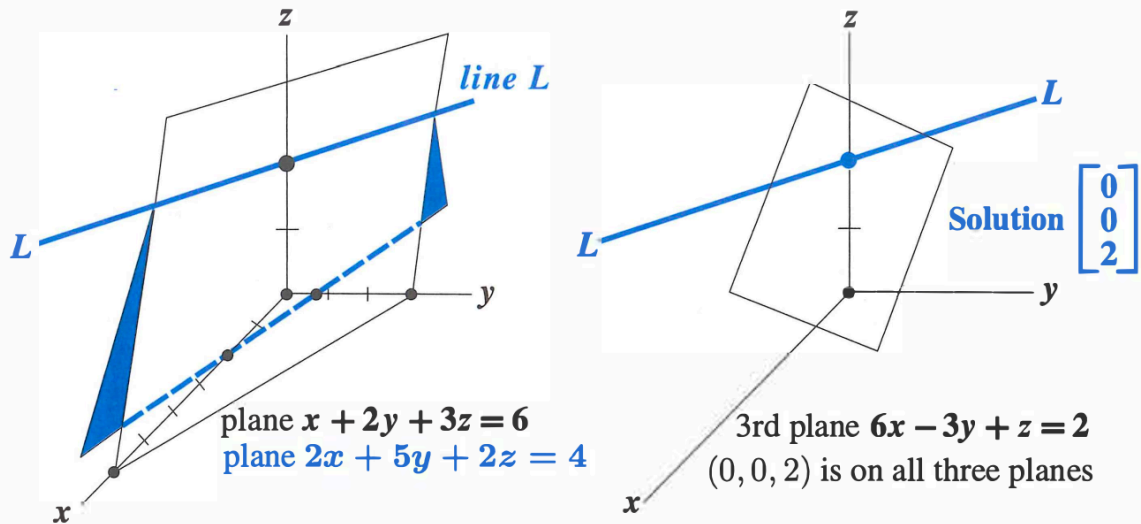


Figure 2.3: Row picture: Two planes meet at a line L . Three planes meet at a point.

with the column picture starts with the vector form of the equations $A\vec{x} = \vec{b}$:

$$\text{Combine Columns} \quad x \cdot \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix} + y \cdot \begin{bmatrix} 2 \\ 5 \\ -3 \end{bmatrix} + z \cdot \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix} = \vec{b}$$

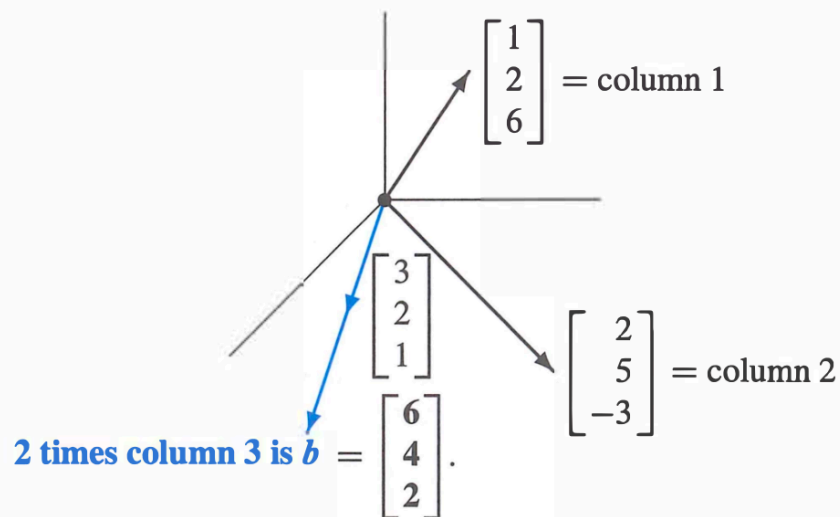


Figure 2.4: Column picture: Combine the columns with weights $(x, y, z) = (0, 0, 2)$.

2. The matrix form of the Equations

For the equations :

$$\begin{cases} x + 2y + 3z = 6 \\ 2x + 5y + 2z = 4 \\ 6x - 3y + z = 2 \end{cases}$$

we get the Coefficient matrix :

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 6 & -3 & 1 \end{bmatrix}$$

thus, we have the Matrix Equation :

$$A \mathbf{x} = \mathbf{b} : \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 6 & -3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix}$$

• What' the mean to “Multiply A times $\vec{x}$ ” ?

• *The Row view :*

$A\vec{x}$ comes from **dot products** :

$$A\vec{x} = \begin{bmatrix} (\text{row } 1) \cdot \vec{x} \\ (\text{row } 2) \cdot \vec{x} \\ (\text{row } 3) \cdot \vec{x} \end{bmatrix}$$

• *The column View :*

$A\vec{x}$ is a combination of column vectors :

$$A\vec{x} = x \cdot (\text{column } 1) + y \cdot (\text{column } 2) + z \cdot (\text{column } 3)$$

3. The Matrix Notation

For a matrix :

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

, the entry $a_{57} = A(5, 7)$ would be in the row 5, column 7. so, above matrix can be denoted as :

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} A(1, 1) & A(1, 2) \\ A(2, 1) & A(2, 2) \end{bmatrix}$$

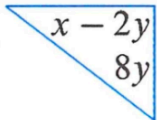
4. The idea of Elimination

Before elimination, x and y appear in both equations. after elimination, the first unknown x has disappeared from the second equation $8y = 8$

Before

$$\begin{array}{r} x - 2y = 1 \\ 3x + 2y = 11 \end{array}$$

After



$$\begin{array}{r} x - 2y = 1 \\ 8y = 8 \end{array}$$

(multiply equation 1 by 3)
(subtract to eliminate 3x)

By this form, we can instantly get the $y = 1$ from $8y = 8$.

- **Elimination** produce the **Upper triangular system** this is our goal. because in the bottom of this system we can instantly get the basic solution (in this instance is $y = 1$).
- The system solved this equations from bottom upwards — first we $y = 1$ and then $x = 3$. this process called '**Back substitution**'

In the next, we will introduce the “Pivot” and “Multiplier” :

Assume :

$$\begin{cases} 4x - 8y = 4 \\ 3x + 2y = 11 \end{cases}$$

we get below result:

$$\begin{cases} 4x - 8y = 4 \\ 8y = 8 \end{cases}$$

by Operations :

Multiply equation 1 by $\frac{3}{4}$
Subtract from equation 2

- **Pivot** : the first nonzero in the row that does the elimination.
- **Multiplier** : (entry to elimination) divided by (pivot) = $\frac{3}{4}$.so the Multiplier is

$$\text{Multiplier} = \frac{\text{Entry to eliminate in row } i}{\text{Pivot in row } j}$$

thus, **Pivots can't be zero !!!**

the new equations with the second pivot is 8, we would use it to eliminate y from the third equation if there were one.

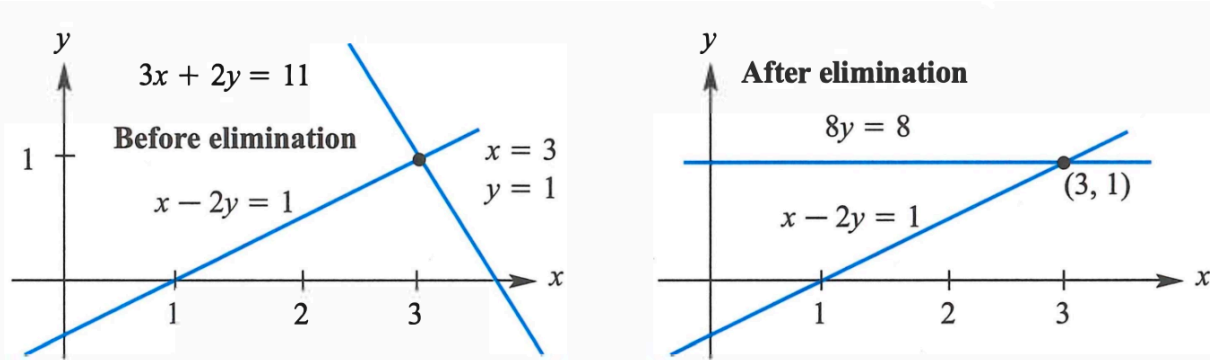


Figure 2.5: Eliminating x makes the second line horizontal. Then $8y = 8$ gives $y = 1$.

4.1. Breakdown of Elimination

Normally, elimination produces the pivots that take us to the solution. But in some situation, the Elimination maybe failure. At this point, the method (elimination) might ask us to divide by zero. Look below examples :

> 1. Example 1 : Fails with no solution to

$$0y = 8$$

. Below figure shows the Row picture and the Column Picture for the Example 1: no solution

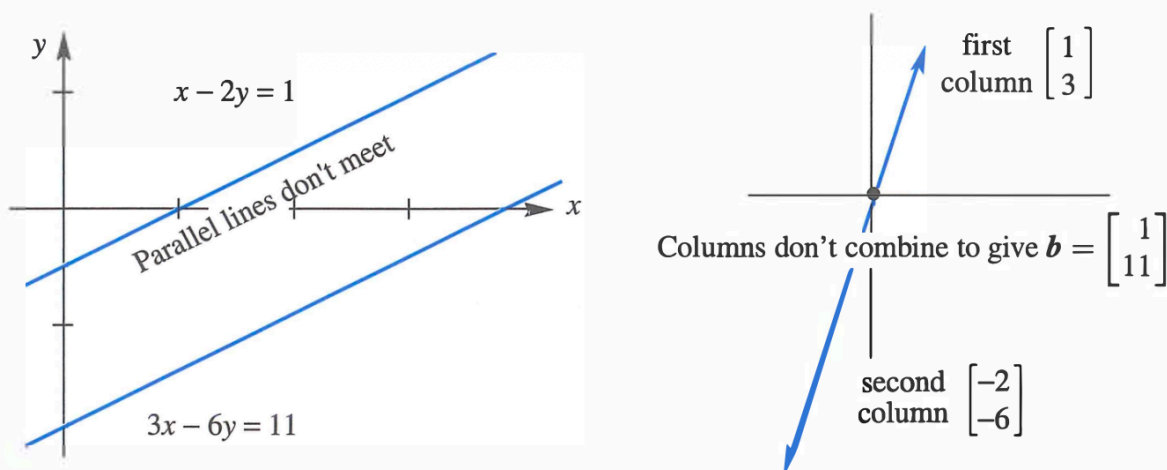


Figure 2.6: Row picture and column picture for Example 1: **no solution**.

Permanent with no solution :

$$\begin{cases} x - 2y = 1 \\ 3x - 6y = 11 \end{cases} \Rightarrow \text{Elimination} \Rightarrow \begin{cases} x - 2y = 1 \\ 0y = 8 \end{cases}$$

there is no solution to $0y = 8$. **Zero is never allowed as a pivot !!!** Here two views about why the pivot can't be zero :

- **Row Picture** : two lines are failure to meet at one point.
- **Column Picture** : two vectors can't be combined to produce the third vector $\vec{b}$.

> 2. Example 2 : Fails with too many solutions to $0y = 0$

$$\begin{cases} x - 2y = 1 \\ 3x - 6y = 3 \end{cases} \Rightarrow \text{Elimination} \Rightarrow \begin{cases} x - 2y = 1 \\ 0y = 0 \end{cases}$$

- **Row picture** : All solutions to this equations are on the same line.
- **Column Picture** : the right side vector $\vec{b} = (1, 3)$ is same as column 1.

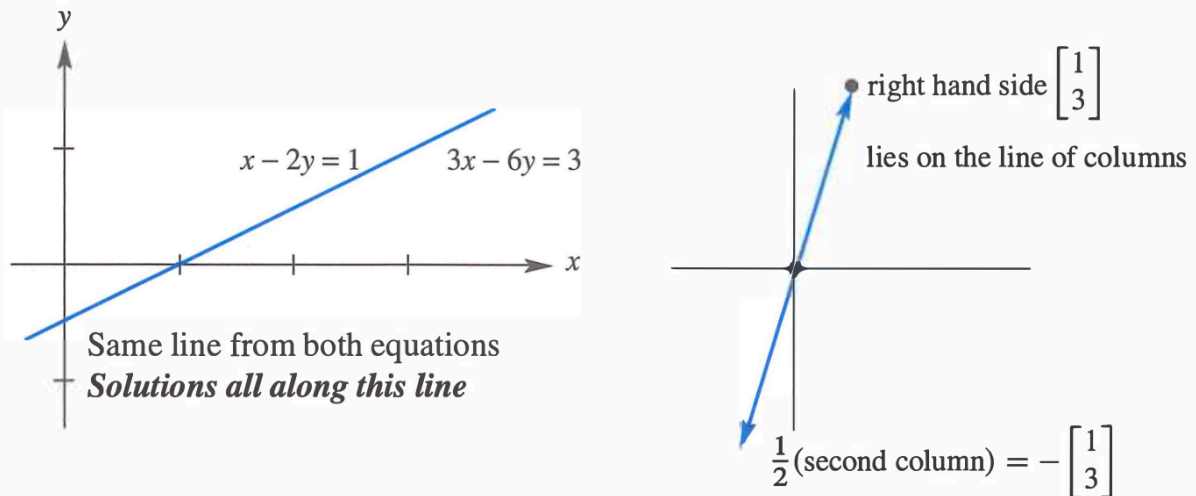


Figure 2.7: Row and column pictures for Example 2: *infinitely many solutions*.

- **Failure** : for n equations we do not get n pivots.
- **Elimination leads to an equation** $0 \neq 0$ (no solution) or $0 = 0$ (many solutions).

> 3. Example 3 :

succeeds by exchanging the equations.

Success comes with n pivots. But we may have to exchange the n equations. :

- If the first pivot position contains zero, we can exchange it with other nonzero elements :

$$\begin{cases} 0x + 2y = 4 \\ 3x - 2y = 5 \end{cases} \Rightarrow \text{Exchange the equation 1 with equation 2}$$

$$\Rightarrow \begin{cases} 3x - 2y = 5 \\ 2y = 4 \end{cases} \Rightarrow \begin{cases} y = 2 \\ x = 3 \end{cases}$$

the new equations system is already triangular. $\begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix}$

4.2. Three Equations in Three Unknowns

$$\begin{cases} 2x + 4y - 2z = 2 \\ 4x + 9y - 3z = 8 \\ -2x - 3y + 7z = 10 \end{cases}$$

we would eliminate the Unknowns step by step :

- **Step 1** : To eliminate the unknown x : we subtract 2 times equation 1 from equation 2. this leads to

$$y + z = 4$$

. the result is :

$$\begin{cases} 2x + 4y - 2z = 2 \\ y + z = 4 \\ -2x - 3y + 7z = 10 \end{cases}$$

- **Step 2** : Subtract 1 times equation 1 from equation 3 to produce

$$y + 5z = 12$$

and

$$\begin{cases} 2x + 4y - 2z = 2 \\ y + z = 4 \\ y + 5z = 12 \end{cases}$$

- **Step 3** : Subtract the equation 2 from 3 :

$$\begin{cases} 2x + 4y - 2z = 2 \\ y + z = 4 \\ 4z = 8 \end{cases}$$

So, we finally get the upper triangular U :

$$U\vec{x} = \vec{c}$$

if we note it as matrix form :

$$\underbrace{\begin{bmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 4 \end{bmatrix}}_U \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}}_c$$

. Notice the pivots 2, 1, 4 along the diagonal of U .

by above equations we can quickly get the solution to this equations :

$$\begin{cases} x = -1 \\ y = 2 \\ z = 3 \end{cases}$$

so , the final form will be :

$$\underbrace{\begin{bmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 4 \end{bmatrix}}_U \underbrace{\begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}}_x = (-1) \cdot \begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix} + 2 \cdot \begin{bmatrix} 4 \\ 9 \\ -3 \end{bmatrix} + 2 \cdot \begin{bmatrix} -2 \\ -3 \\ 7 \end{bmatrix} = \underbrace{\begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}}_c$$

5. Elimination Using Matrices

suppose we have a euqations :

$$\begin{cases} 2x_1 + 4x_2 - 2x_3 = 2 \\ 4x_1 + 9x_2 - 3x_3 = 8 \\ -2x_1 - 3x_2 + 7x_3 = 10 \end{cases}$$

is same as

$$\begin{bmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 8 \\ 10 \end{bmatrix} \Rightarrow A\vec{x} = \vec{b}$$

When the number of equations (three) mathcs the number of the Unknowns (three). Our matrix is 3 by 3.

• **The number of row of the matrix must mathcs the number of the Unknowns.**

To solve this Equations we use the Elimination but is Matrix Form :

• **Step 1 :** To eliminate the $e_{(2,1)}$ in the original equations :

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{E_{(2,1)}} \underbrace{\begin{bmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{bmatrix}}_A \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 8 \\ 10 \end{bmatrix} \Rightarrow$$

$$\begin{bmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ -2 & -3 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 10 \end{bmatrix}$$

• **Step 2 :** To eliminate the $e_{(3,1)}$:

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}}_{E_{(3,1)}} \begin{bmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ -2 & -3 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 10 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ 0 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 12 \end{bmatrix}$$

• **Step 3 :**

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}}_{E_{(3,2)}} \begin{bmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ 0 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 12 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 4 \end{bmatrix}}_U \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}$$

so we get the Elimination Matrix

$$E = E_{(2,1)} E_{(3,1)} E_{(3,2)} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$$

thus, we can get :

$$A\vec{x} = b \Rightarrow \\ EA\vec{x} = E\vec{b} \Rightarrow U\vec{x} = E\vec{b}$$

6. Augmented Matrix

Suppose we have two matrix $A : \begin{bmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{bmatrix}$ and $b : \begin{bmatrix} 2 \\ 8 \\ 10 \end{bmatrix}$ we can put them together to produce the “**Augmented Matrix**” :

$$\text{Augmented Matrix } [A \ b] = \begin{bmatrix} 2 & 4 & -2 & 2 \\ 4 & 9 & -3 & 8 \\ -2 & -3 & 7 & 10 \end{bmatrix}$$

We can execute the elimination at the same time with Augmented Matrix :

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & -2 & 2 \\ 4 & 9 & -3 & 8 \\ -2 & -3 & 7 & 10 \end{bmatrix} = \begin{bmatrix} 2 & 4 & -2 & 2 \\ 0 & 1 & 1 & 4 \\ -2 & -3 & 7 & 10 \end{bmatrix}$$

7. The Matrix P_{ij} for a Row Exchange

The P_{ij} Matrix is also known “Permutation Matrix”. A row exchange is needed when zero is in the pivot position.

First we introduce the P_{23} , it exchanges the row 2 and row 3 :

$$P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Such as :

$$P_{23} \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 3 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 4 & 1 \\ 0 & 0 & 3 \\ 0 & 6 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 1 \\ 0 & 6 & 5 \\ 0 & 0 & 3 \end{bmatrix}$$

• **Row Exchange Matrix** : P_{ij} is the identity matrix with row i and j reversed.

8. Indetity Matrix

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

9. Rules for Matrix Operations

9.1. Matrix Multiplication

Fundamental Law of Matrix Multiplication :

1. $A_{m \times n} B_{n \times p} = C_{m \times p}$
2. $(AB)C = A(BC)$
3. The entry in row i and column j of AB :
 $(\text{row } i \text{ of } A) \times (\text{column } j \text{ of } B)$

Figure 2.8 picks out the second row ($i = 2$) of a 4 by 5 matrix A . It picks out the third column ($j = 3$) of a 5 by 6 matrix B . Their dot product goes into row 2 and column 3 of AB . The matrix AB has *as many rows as* A (4 rows), and *as many columns as* B .

$$\begin{bmatrix} * \\ a_{i1} & a_{i2} & \cdots & a_{i5} \\ * \\ * \end{bmatrix} \begin{bmatrix} * & * & b_{1j} & * & * & * \\ & & b_{2j} & & & \\ & & \vdots & & & \\ & & b_{5j} & & & \end{bmatrix} = \begin{bmatrix} * & * & * & * & * & * \\ * & (AB)_{ij} & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \end{bmatrix}$$

A is 4 by 5 B is 5 by 6 AB is $(4 \times 5)(5 \times 6) = 4$ by 6

Figure 2.8: Here $i=2$ and $j=3$. Then $(AB)_{23}$ is **(row 2) · (column 3)** = sum of $a_{2k}b_{k3}$.

Suppose we have two Matrices :

$$E : \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad A : \begin{bmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{bmatrix}$$

So the Multiplication E by A is :

$$\begin{aligned} EA &= \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 9 \\ -3 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ -3 \\ 7 \end{bmatrix} \end{aligned}$$

so ,

$$\text{Matrix Multiplication } AB = A[\vec{b}_1 \vec{b}_2 \vec{b}_3] = [A\vec{b}_1 A\vec{b}_2 A\vec{b}_3]$$

9.1.1. The Columns view : A linear combination to the matrix A

Matrix A times every column of B :

$$A[b_1, \dots, b_p] = [Ab_1, \dots, Ab_p]$$

Each column of AB is a **combination of the columns of A** .

9.1.2. The Rows view : A linear combination to the Matrix B

Every row of A times matrix B :

$$[\text{row } i \text{ of } A]B = [\text{row } i \text{ of } AB]$$

Every row of AB is a **combination of the rows of B** .

9.1.3. Columns Multiply Rows

$$\begin{aligned} AB &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} E & F \\ G & H \end{bmatrix} = \begin{bmatrix} aE + bG & aF + bH \\ cE + dG & cF + dH \end{bmatrix} \\ &\Rightarrow \\ &\begin{bmatrix} a \\ c \end{bmatrix} [E \ F] + \begin{bmatrix} b \\ d \end{bmatrix} [G \ H] \end{aligned}$$

9.2. Basic Rules :

1. $A + B = B + A$
2. $c(A + B) = cA + cB$
3. $A + (B + C) = (A + B) + C$
4. $AB \neq BA$
5. $IA = AI$

with the I is "Identity Matrix"

9.3. Block Matrices and Block Multiplication

Any Matrices are can cut into blocks. Here is a 4 by 6 matrix broken into blocks of size of 2 by 2 :

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} I & I & I \\ I & I & I \end{pmatrix}.$$

- If B also is 4 by 6 and the block sizes match, we can add $A + B$ a block at a time.
- If blocks of A can multiply blocks of B , then block multiplication of AB is allowed. Cuts between columns of A match cuts between rows of B

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} \\ B_{21} \end{bmatrix} = \begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix} [B_{11}] + \begin{bmatrix} A_{12} \\ A_{22} \end{bmatrix} [B_{21}]$$

$$= \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} \\ A_{21}B_{11} + A_{22}B_{21} \end{bmatrix}$$

- Important Special Case :

$$\begin{array}{l} \text{Columns} \\ \text{times} \\ \text{rows} \end{array} \quad \begin{bmatrix} | & & | \\ \mathbf{a}_1 & \cdots & \mathbf{a}_n \\ | & & | \end{bmatrix} \begin{bmatrix} - & \mathbf{b}_1 & - \\ & \vdots & \\ - & \mathbf{b}_n & - \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1 \mathbf{b}_1 + \cdots + \mathbf{a}_n \mathbf{b}_n \end{bmatrix}. \quad (5)$$

9.3.1. Block Elimination

Suppose a matrix has four blocks :

$$A = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

we will construct a block matrix to eliminate the block C within the matrix A :

$$\begin{aligned} \text{Block Multiplication : } \begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} &= \begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \end{bmatrix} + \begin{bmatrix} 0 \\ I \end{bmatrix} \begin{bmatrix} C & D \end{bmatrix} \\ &= \begin{bmatrix} A & B \\ -CA^{-1}A & -CA^{-1}B \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ C & D \end{bmatrix} \\ &= \begin{bmatrix} A & B \\ 0 & D - CA^{-1}B \end{bmatrix} \end{aligned}$$

by this process we get the “**Schur Complement**”:

$$S = D - CA^{-1}B$$